

ECEN 662:

Estimation & Detection Theory

Lecture 6: Uniformly Minimum-Variance Unbiased Estimation (Part III)

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- During the previous lecture, we discussed how to “certify” the optimality of an unbiased estimator based on *unbiased estimators of 0*
- In this lecture, we shall introduce a *universal* lower bound on the variance of *any* unbiased estimator
- Therefore, any unbiased estimator whose variance *matches* this universal lower bound is the UMVUE

Fisher information

- Assume that our likelihood function $f_\theta(x)$ satisfies the following two *regularity* assumptions:
 - The *support* $\{x : f_\theta(x) > 0\}$ does *not* depend on θ
 - For *any* scalar statistic $T(X)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$,

$$\nabla_\theta (\mathbb{E}_\theta [T(X)]) = \int_{\mathcal{X}} T(x) \nabla_\theta (f_\theta(x)) dx$$

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- (*The fundamental identity of score function*)

$$\mathbb{E}_\theta[s_\theta(X)T(X)] = \nabla_\theta (\mathbb{E}_\theta[T(X)])$$

for any scalar statistic $T(X)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$

Proof of the score identity

$$\begin{aligned}\mathbb{E}_\theta[s_\theta(X)T(X)] &= \int_{\mathcal{X}} s_\theta(x)T(x)f_\theta(x)dx \\ &= \int_{\mathcal{X}} \nabla_\theta (\log f_\theta(x)) T(x)f_\theta(x)dx \\ &= \int_{\mathcal{X}} \nabla_\theta (f_\theta(x)) T(x)dx \\ &= \nabla_\theta \left(\int_{\mathcal{X}} f_\theta(x)T(x)dx \right) \\ &= \nabla_\theta (\mathbb{E}_\theta[T(X)])\end{aligned}$$

Fisher information and average curvature

- Applying the score identity with $T(x) = 1$ gives

$$\mathbb{E}_\theta[s_\theta(X)] = \nabla_\theta (\mathbb{E}_\theta[1]) = 0,$$

i.e., the score function $s_\theta(X)$ always has a *zero* mean

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- The *covariance matrix* of the score function

$$\text{Var}_\theta[s_\theta(X)] = \mathbb{E}_\theta[s_\theta(X)s_\theta^t(X)]$$

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- If the log-likelihood $\log f_\theta(x)$ is *twice* differentiable with respect to θ and under the additional regularity assumption that for any scalar statistic $T(x)$ such that $\mathbb{E}_\theta[|T(x)|] < \infty$ for all $\theta \in \Theta$,

$$\nabla_\theta^2 (\mathbb{E}_\theta [T(x)]) = \int_{\mathcal{X}} T(x) \nabla_\theta^2 (f_\theta(x)) dx$$

- We have

$$\nabla_{\theta}^2 (\log f_{\theta}(x)) = \frac{\nabla_{\theta}^2 (f_{\theta}(x))}{f_{\theta}(x)} - s_{\theta}(x) s_{\theta}^t(x)$$

and hence

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- Note that $-\frac{\partial^2}{\partial \theta^2} \log f_{\theta}(x)$ is the *curvature* of the log-likelihood $\log f_{\theta}(x)$ with respect to θ for a given x , so Fisher information measures the *average* curvature of the log-likelihood $\log f_{\theta}(x)$ at a given value of θ

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- More specifically, let $s_\eta(x)$ and $I(\eta)$ be Fisher score and the Fisher information with respect to the parameter η , respectively
- If $\eta = \eta(\theta)$ is a *continuous and differentiable* mapping from θ to η , we have

$$s_\theta(x) = \nabla_\theta \left(\log \tilde{f}_\eta(x) \right) = J_\eta^t \nabla_\eta \left(\log \tilde{f}_\eta(x) \right) = J_\eta^t s_\eta(x)$$

where $J_\eta = \left[\frac{\partial \eta_i}{\partial \theta_j} \right]$ is the *Jacobian* of $\eta = \eta(\theta)$

- We thus have the following *re-parameterization* formula for Fisher information:

$$\begin{aligned} I(\theta) &= \mathbb{E}_\theta[s_\theta(X)s_\theta^t(X)] \\ &= J_\eta^t \mathbb{E}_\theta[s_\eta(X)s_\eta^t(X)] J_\eta \\ &= J_\eta^t \mathbb{E}_\eta[s_\eta(X)s_\eta^t(X)] J_\eta \\ &= J_\eta^t I(\eta) J_\eta \end{aligned}$$

Exponential family of distributions

- In particular, an *exponential* family of distributions can be re-parameterized in terms of the natural parameter η as:

$$\tilde{f}_{\eta}(x) = h(x) \exp(\eta^t T(x) - A(\eta))$$

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- Note that the score function with respect to the natural parameter can be calculated as:

$$\begin{aligned} s_\eta(x) &= \nabla_\eta \log \tilde{f}_\eta(x) = T(x) - \nabla_\eta A(\eta) \\ \text{and } \nabla_\eta^2 \log \tilde{f}_\eta(x) &= -\nabla_\eta^2 A(\eta) \end{aligned}$$

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- By the *re-parameterization* formula, the Fisher information with respect to θ is given by

$$I(\theta) = J_\eta^t I(\eta) J_\eta = J_\eta^t \nabla_\eta^2 A(\eta) J_\eta$$

Additivity over independent measurements

- Let $X = (X_1, \dots, X_n)$ be an observation of n *independent* measurements:

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- By the *independence* of the measurements, the Fisher information

$$I_n(\theta) = \text{Var}_{\theta} [s_{\theta}(X)] = \sum_{i=1}^n \text{Var}_{\theta} [s_{\theta}(X_i)] = nI_1(\theta)$$

i.e., Fisher information is *additive* over independent measurements

Example: Gaussian with unknown mean and variance

- Recall that in this case ($\theta_1 = \mu$ and $\theta_2 = \sigma^2$), the marginal distributions are exponential families of distributions with

$$\eta(\theta) = \left(\frac{\theta_1}{\theta_2}, -\frac{1}{2\theta_2} \right), \quad T(x_i) = (x_i, x_i^2)$$

and log-partition function

$$A(\eta) = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2} \log(-\eta_2)$$

- The Fisher information

$$\begin{aligned}
I_n(\theta) &= nI_1(\theta) \\
&= nJ_\eta^t \nabla_\eta^2 A(\eta) J_\eta \\
&= n \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix}^t \begin{bmatrix} -\frac{1}{2\eta_2} & \frac{\eta_1}{2\eta_2^2} \\ \frac{\eta_1}{2\eta_2^2} & -\frac{\eta_1^2}{2\eta_2^3} + \frac{1}{2\eta_2^2} \end{bmatrix} \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix} \\
&= n \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix}^t \begin{bmatrix} \theta_2 & 2\theta_1\theta_2 \\ 2\theta_1\theta_2 & 4\theta_1^2\theta_2 + 2\theta_2^2 \end{bmatrix} \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix} \\
&= \begin{bmatrix} \frac{n}{\theta_2} & 0 \\ 0 & \frac{n}{2\theta_2^2} \end{bmatrix} \\
&= \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}
\end{aligned}$$

Cramér-Rao lower bound (CRLB)

- Assume that the likelihood function $f_\theta(x)$ satisfies the aforementioned *regularity* assumptions. We have

$$\text{Var}_\theta[\hat{g}(X)] \geq J_\phi I^{-1}(\theta) J_\phi^t, \quad \forall \theta \in \Theta$$

for any estimator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)],$$

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- Therefore, any *unbiased* estimator whose variance *equals*

$$J_g I^{-1}(\theta) J_g^t$$

is the *UMVUE* for $g(\theta)$

Proof

- For simplicity, let us assume that θ is a *scalar* parameter. The vector case follows essentially the same idea, but requires a bit more machinery

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- By the *Cauchy-Schwarz inequality*, we have

$$\text{Var}_\theta[s_\theta(X)]\text{Var}_\theta[\hat{g}(X)] \geq (\text{Cov}_\theta[s_\theta(X), \hat{g}(X)])^2$$

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- By the score identity, the covariance

$$\begin{aligned}\text{Cov}_\theta[s_\theta(X), \hat{g}(X)] &= \mathbb{E}_\theta[s_\theta(X)\hat{g}(X)] - \mathbb{E}_\theta[s_\theta(X)]\mathbb{E}_\theta[\hat{g}(X)] \\ &= \mathbb{E}_\theta[s_\theta(X)\hat{g}(X)] \\ &= \frac{d}{d\theta}\mathbb{E}_\theta[\hat{g}(X)] \\ &= \frac{d}{d\theta}\phi(\theta)\end{aligned}$$

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- It follows that

$$\text{Var}_\theta[\hat{g}(X)] \geq \frac{\left(\frac{d}{d\theta}g(\theta)\right)^2}{I(\theta)}$$

for all $\theta \in \Theta$

Example: Gaussian with unknown mean and variance

- Recall that $\theta = (\mu, \sigma^2)$, and the Fisher information matrix is given by:

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- Thus, the CRLB for $g(\theta) = \theta_1 = \mu$ is

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{\sigma^2}{n},$$

and the CRLB for $g(\theta) = \theta_2 = \sigma^2$ is

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{2\sigma^4}{n}$$

- Recall that the *empirical mean*

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$$

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- Also recall that for $n \geq 2$,

$$\widehat{\sigma^2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

is an unbiased estimator of σ^2 . However, its variance is $\frac{2\sigma^4}{n-1}$, which is strictly greater than the CRLB $\frac{2\sigma^4}{n}$

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- Therefore, for σ^2 , we *cannot* conclude from the CRLB that $\widehat{\sigma^2}$ is the UMVUE (even though it, in fact, is)

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- Also keep in mind that Fisher information and, by extension, CRLB are *only* defined for likelihood families that satisfy the *regularity* conditions
- Next, we shall briefly mention the *Hammersley-Chapman-Robbins lower bound (HCRLB)*, which can be applied *without* the regularity conditions

Hammersley-Chapman-Robbins lower bound (HCRLB)

- We have

$$\mathrm{Var}_{\theta}[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{(\phi(\theta') - \phi(\theta))^2}{\mathbb{E}_{\theta} \left[\left(\frac{f_{\theta'}(X)}{f_{\theta}(X)} - 1 \right)^2 \right]}, \quad \forall \theta \in \Theta$$

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- In particular, if $\hat{g}(X)$ is *unbiased*, we have $\phi(\theta) = g(\theta)$ and hence

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{(g(\theta') - g(\theta))^2}{\mathbb{E}_\theta \left[\left(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1 \right)^2 \right]}, \quad \forall \theta \in \Theta$$

- Let $\theta' = \theta + \epsilon$. Under the *regularity* conditions, it can be shown that

$$\lim_{\|\epsilon\| \rightarrow 0} \frac{(\phi(\theta') - \phi(\theta))^2}{\mathbb{E}_{\theta} \left[\left(\frac{f_{\theta'}(X)}{f_{\theta}(X)} - 1 \right)^2 \right]} = J_{\phi} I^{-1}(\theta) J_{\phi}^t, \quad \forall \theta \in \Theta$$

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- The real power of the HCRLB, however, lies in the fact that it can be applied even when the regularity conditions are *not* met

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- Note that for this example, the support of $f_{\theta}(x)$ depends on θ . Therefore, the regularity conditions are *not* met for this likelihood family and the CRLB does not apply

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- This gives

$$\text{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' > \theta} \frac{(\theta' - \theta)^2}{\left(\frac{\theta'}{\theta}\right)^n - 1} = \sup_{\epsilon > 0} \frac{\theta^2 \epsilon^2}{(1 + \epsilon)^n - 1}$$